INFRINGEMENT REPORT

SUBJECT PATENT: **US10298999B2**

PRIORITY DATE: **SEPTEMBER 09, 2014**

INFRINGING ENTITY NAME: **AMAZON**

CHART STRENGTH: **HIGH**

PATENT TECHNOLOGY: **MEDIA CONTENT RULE COMPLIANCE**

PRODUCT NAME: **AMAZON REKOGNITION**

AMAZON REVENUE: **<$650.313 BILLION (SOURCE)**

PRODUCT RELEASE DATE: **2016 (SOURCE)**

US10298999B2	Amazon Rekognition
12. **A method for analyzing content in a server device,** comprising:	*A method (e.g., Amazon Rekognition) for analysing content (e.g., images and videos) in a server device (e.g., Amazon S3).* **Amazon Rekognition** Amazon Rekognition is a cloud-based image and video analysis service that makes it easy to add advanced computer vision capabilities to your applications. The service is powered by proven deep learning technology and it requires no machine learning expertise to use. Amazon Rekognition includes a simple, easy-to-use API that can quickly analyze any image or video file that's stored in Amazon S3. You can add features that detect objects, text, unsafe content, analyze images/videos, and compare faces to your application using Rekognition's APIs. With Amazon Rekognition's face recognition APIs, you can detect, analyze, and compare faces for a wide variety of use cases, including user verification, cataloging, people counting, and public safety. Source: https://docs.aws.amazon.com/pdfs/rekognition/latest/dg/rekognition-dg.pdf#what-is
receiving a digital data stream comprising a plurality of pieces of digital content, each piece of digital content being associated with asset metadata describing the piece of digital content;	*The method further comprises receiving a digital data stream comprising a plurality of pieces of digital content (e.g., images and videos), each piece of digital content being associated with asset metadata (e.g., "Face metadata", "video metadata", etc.) describing the piece of digital content.* **Understanding Rekognition's image and video operations** Amazon Rekognition offers two primary API sets for image and video analysis: • Amazon Rekognition Image: This API is designed for analyzing images. • Amazon Rekognition Video: This API focuses on analyzing both stored and streaming videos.

Amazon Rekognition Image operations

Amazon Rekognition Image operations are synchronous. The input and response are in JSON format. Amazon Rekognition Image operations analyze an input image that is in .jpg or .png image format. The image passed to an Amazon Rekognition Image operation can be stored in an Amazon S3 bucket. If you are not using the AWS CLI, you can also pass Base64 encoded images bytes directly to an Amazon Rekognition operation. For more information, see Working with images.

Amazon Rekognition Video operations

The Amazon Rekognition Video API facilitates the analysis of videos either stored in an Amazon S3 bucket or streamed via Amazon Kinesis Video Streams.

For stored video operations, note the following:

- Operations are asynchronous.
- Analysis must be initiated with a "Start" operation (e.g., StartFaceDetection for face detection in stored videos).
- The completion status of analysis is published to an Amazon SNS topic.
- To retrieve the results of an analysis, use the corresponding "Get" operation (e.g., GetFaceDetection).
- For more information, see Working with stored video analysis.

For streaming video analysis:

- Capabilities include face search in Rekognition Video collections and label (object or concept) detection.
- Analysis results for labels are sent as Amazon SNS and Amazon S3 notifications.
- Face search results are output to a Kinesis data stream.
- Management of streaming video analysis is done via an Amazon Rekognition Video stream processor (e.g., create a processor using CreateStreamProcessor).
- For more information, see Working with streaming video events.

Each video analysis operation returns metadata about the video being analyzed, as well as a job ID and a job tag. Operations like Label Detection and Content Moderation for video allow sorting by timestamp or label name, and aggregating results by timestamp or by segment.

Example 2: An application to display enhanced face images

First, you can detect faces in each of the images in your local repository using the Amazon Rekognition `DetectFaces` operation and build a client-side index. For each face, the operation returns metadata that includes a bounding box, facial landmarks (for example, the position of mouth and ear), and facial attributes (for example, gender). You can store this metadata in a client-side local index, as shown following:

```
ImageID     FaceID     FaceMetaData

image-1     face-1     <boundingbox>, etc.
image-1     face-2     <boundingbox>, etc.
image-1     face-3     <boundingbox>, etc.
...
```

Getting Amazon Rekognition Video analysis results

To get the results of a video analysis request, first ensure that the completion status that's retrieved from the Amazon SNS topic is SUCCEEDED. Then call `GetLabelDetection`, which passes the `JobId` value that's returned from `StartLabelDetection`. The request JSON is similar to the following example:

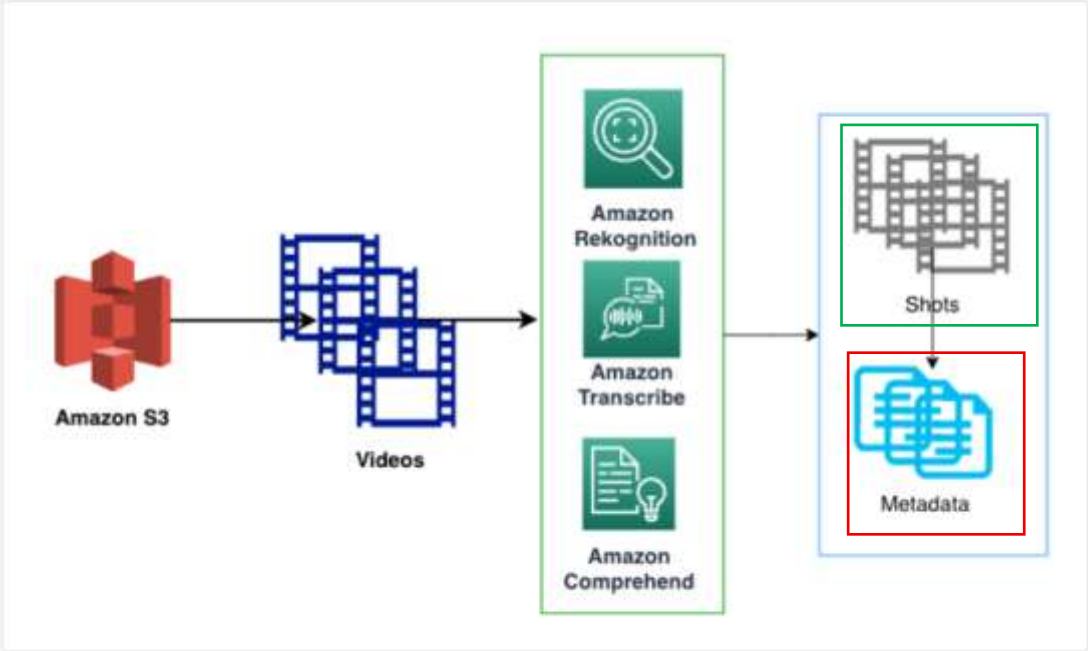
```
{
    "JobId": "270c1cc5e1d0ea2fbc59d97cb69a72a5495da75851976b14a1784ca90fc180e3",
    "MaxResults": 10,
    "SortBy": "TIMESTAMP"
}
```

```
    ],
    "LabelModelVersion": "2.0",
    "JobStatus": "SUCCEEDED",
    "VideoMetadata": {
        "Format": "QuickTime / MOV",
        "FrameRate": 23.976024627685547,
        "Codec": "h264",
        "DurationMillis": 5005,
        "FrameHeight": 674,
        "FrameWidth": 1280
    }
}
```

Source: https://docs.aws.amazon.com/pdfs/rekognition/latest/dg/rekognition-dg.pdf#what-is

Metadata generation pipeline

The video metadata generation pipeline consists of processing video files using AWS services such as Amazon Transcribe, Amazon Rekognition, and Amazon Comprehend, as shown in the following diagram. The metadata is generated at the shot level for a video.



In this section, we discuss the details of each service and the workflow in more detail.

The solution runs Amazon Rekognition APIs for label detection, text detection, celebrity detection, and face detection on videos. The metadata generated for each video by the APIs is processed and stored with timestamps. The videos are then segmented into individual shots. With Amazon Rekognition, you can detect the start, end, and duration of each shot as well as the total shot count for a content piece. The video shot detection job starts with the `StartSegmentDetection` API, which returns a `jobId` that can be used to monitor status with the `GetSegmentDetection` API. When the video segmentation status changes to `Succeeded`, for each shot, you parse the previously generated Amazon Rekognition API metadata using the shot's timestamp. You then append this parsed metadata to the shot record. Similarly, the full transcript from Amazon Transcribe is segmented using the shot start and end timestamps to create shot-level transcripts.

	The processing algorithm is as follows: rekognition_metadata - shot_metadata: extract StartFrameNumber and EndFrameNumber - celeb_metadata: extract celeb_metadata - label_metadata: extract unique labels - text_metadata: extract unique text labels if there are more than 3 words (comes noisy with "-", "null" and other values) - face_analysis_metadata: extract unique list of AgeRange, Emotions, Gender We combine all rekognition text data into 'rek_text_metadata' string transcribe_metadata - transcribe_metadata: check the wordcount of the conversation across all speakers. if it is more than 50 words, mark it for summarization task with Amazon Bedrock comprehend_metadata - comprehend_metadata: extract sentiment - comprehend_metadata: extract target sentiment scores for words with score > 0.9 Source: https://aws.amazon.com/blogs/machine-learning/how-veritone-uses-amazon-bedrock-amazon-rekognition-amazon-transcribe-and-information-retrieval-to-update-their-video-search-pipeline/
accessing a set of compliance rules;	*The method (e.g., Amazon Rekognition) further comprises accessing a set of compliance rules (e.g., compliance validation).* **Compliance validation for Amazon Rekognition** Third-party auditors assess the security and compliance of Amazon Rekognition as part of multiple AWS compliance programs. These include SOC, PCI, FedRAMP, HIPAA, and others. For a list of AWS services in scope of specific compliance programs, see AWS Services in Scope by Compliance Program. For general information, see AWS Compliance Programs. You can download third-party audit reports using AWS Artifact. For more information, see Downloading Reports in AWS Artifact.

	Your compliance responsibility when using Amazon Rekognition is determined by the sensitivity of your data, your company's compliance objectives, and applicable laws and regulations. AWS provides the following resources to help with compliance: • Security and Compliance Quick Start Guides – These deployment guides discuss architectural considerations and provide steps for deploying security- and compliance-focused baseline environments on AWS. • Architecting for HIPAA Security and Compliance Whitepaper – This whitepaper describes how companies can use AWS to create HIPAA-compliant applications. • AWS Compliance Resources – This collection of workbooks and guides might apply to your industry and location. • AWS Config – This AWS service assesses how well your resource configurations comply with internal practices, industry guidelines, and regulations. • AWS Security Hub – This AWS service provides a comprehensive view of your security state within AWS that helps you check your compliance with security industry standards and best practices. Source: https://docs.aws.amazon.com/pdfs/rekognition/latest/dg/rekognition-dg.pdf#what-is  The AWS Compliance Program helps customers to understand the robust controls in place at AWS to maintain security and compliance of the cloud. By tying together governance-focused, audit-friendly service features with applicable compliance or audit standards, AWS Compliance Enablers build on traditional programs, helping customers to establish and operate in an AWS security control environment. Source: https://aws.amazon.com/compliance/programs/
comparing each received piece of digital content with the accessed compliance rules;	*The method (e.g., Amazon Rekognition) further comprises comparing each received piece of digital content (e.g., images and videos) with the accessed compliance rules.*

Analyzing images stored in an Amazon S3 bucket

Amazon Rekognition Image can analyze images that are stored in an Amazon S3 bucket or images that are supplied as image bytes.

In this topic, you use the DetectLabels API operation to detect objects, concepts, and scenes in an image (JPEG or PNG) that's stored in an Amazon S3 bucket. You pass an image to an Amazon Rekognition Image API operation by using the Image input parameter. Within `Image`, you specify the S3Object object property to reference an image stored in an S3 bucket. Image bytes for images stored in Amazon S3 buckets don't need to be base64 encoded. For more information, see Image specifications.

Detection of Personal Protective Equipment - Detect personal protective equipment in images to monitor safety compliance across various industries. You can automatically flag unsafe conditions by detecting improper equipment and receive alerts about these conditions, which can improve compliance and training.

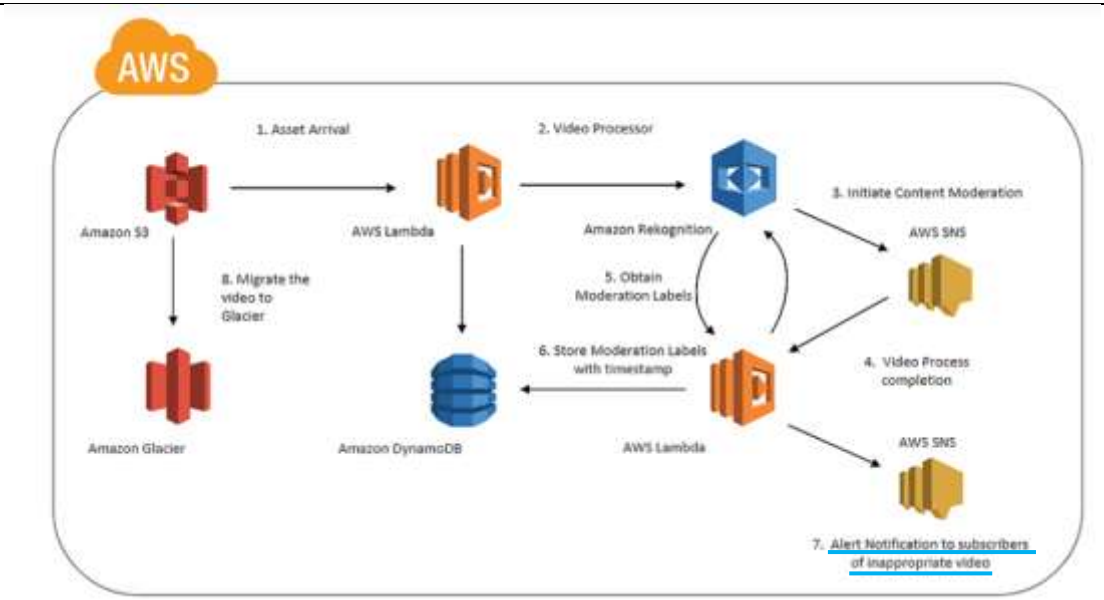
For text detection in videos, you can implement solutions such as:

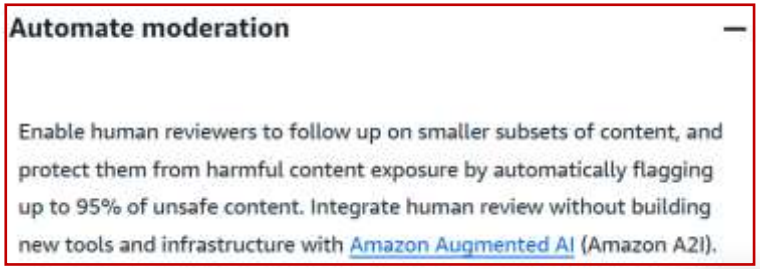
- Searching videos for clips with specific text keywords, such as a guest's name on a graphic in a news show.
- Moderating content for compliance with organizational standards by detecting accidental text, profanity, or spam.
- Finding all text overlays on the video timeline for further processing, such as replacing text with text in another language for content internationalization.
- Finding text locations, so that other graphics can be aligned accordingly.

Using the image and video moderation APIs

In the Amazon Rekognition Image API, you can detect inappropriate, unwanted, or offensive content synchronously using DetectModerationLabels and asynchronously using StartMediaAnalysisJob and GetMediaAnalysisJob operations. You can use the Amazon Rekognition Video API to detect such content asynchronously by using the StartContentModeration and GetContentModeration operations.

	You can use Amazon Rekognition to detect content that is inappropriate, unwanted, or offensive. You can use Rekognition moderation APIs in social media, broadcast media, advertising, and e-commerce situations to create a safer user experience, provide brand safety assurances to advertisers, and comply with local and global regulations. Source: https://docs.aws.amazon.com/pdfs/rekognition/latest/dg/rekognition-dg.pdf#what-is
generating an alert if a result of the comparing step indicates a breach of at least one of the rules;	*The method (e.g., Amazon Rekognition) further comprises generating an alert (e.g., alert notifications) if a result of the comparing step indicates a breach of at least one of the rules (e.g., according to compliance requirements)* **Detection of Personal Protective Equipment** - Detect personal protective equipment in images to monitor safety compliance across various industries. You can automatically flag unsafe conditions by detecting improper equipment and receive alerts about these conditions, which can improve compliance and training. Source: https://docs.aws.amazon.com/pdfs/rekognition/latest/dg/rekognition-dg.pdf#what-is In this blog post, I show you how to build a serverless architecture to evaluate video content for enhanced compliance and moderation using Amazon Rekognition Video. Customers (especially media and entertainment companies) face the challenge of categorizing videos accordingly for building age-appropriate audience content or identifying inappropriate content in videos. Amazon Rekognition Video is a deep-learning-powered video analysis service that follows the path of objects, detects activities, and recognizes objects, celebrities, and content types. Amazon Rekognition Video can detect explicit and questionable content, so that you can filter videos based on your applications and your compliance requirements.

	 Processing starts immediately after media has been uploaded to our asset ingest Amazon S3 bucket. Amazon S3 event notifications triggers a video processor Lambda function that initiates the StartContentModeration API action on the video file. The completion status of media processing by Amazon Rekognition Video is obtained through Amazon SNS, which in turn triggers a content alert Lambda function. This Lambda function obtains the moderation labels by using the GetContentModeration API action, parsing the labels, and alerting through an SNS topic. Teams subscribed to the SNS topic will be immediately alerted with the name of the file and the count of moderation labels present. Complete moderation label information along with a timestamp is stored in the database. The video file in the S3 bucket is sent for archiving to Amazon Glacier, which provides low cost archive storage, and this completes our processing pipeline. Source: https://aws.amazon.com/blogs/machine-learning/build-an-automatic-alert-system-to-easily-moderate-content-at-scale-with-amazon-rekognition-video/
outputting to a moderator, responsive to the alert, an identification associated with only the piece of digital content which caused the alert to be generated;	*The method (e.g., Amazon Rekognition) further comprises outputting to a moderator (e.g., content moderator), responsive to the alert (e.g., alert notification), an identification (e.g., improper equipment, explicit and questionable content etc.) associated with only the piece of digital content (e.g., images or videos etc.) which caused the alert to be generated.*



Source: https://aws.amazon.com/rekognition/content-moderation/

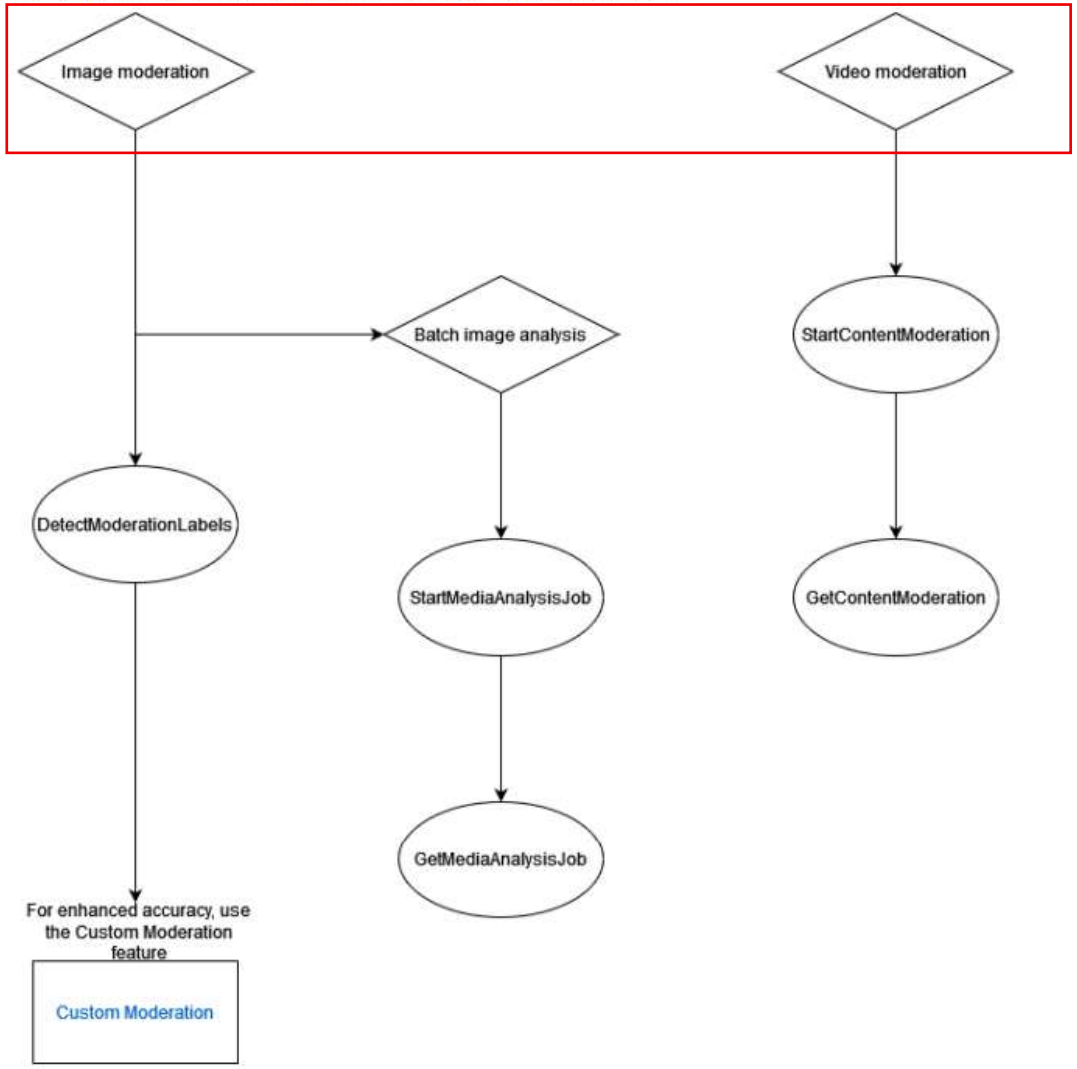
Moderating content

You can use Amazon Rekognition to detect content that is inappropriate, unwanted, or offensive. You can use Rekognition moderation APIs in social media, broadcast media, advertising, and e-commerce situations to create a safer user experience, provide brand safety assurances to advertisers, and comply with local and global regulations.

Today, many companies rely entirely on human moderators to review third-party or user-generated content, while others simply react to user complaints to take down offensive or inappropriate images, ads, or videos. However, human moderators alone cannot scale to meet these needs at sufficient quality or speed, which leads to a poor user experience, high costs to achieve scale, or even a loss of brand reputation. By using Rekognition for image and video moderation, human moderators can review a much smaller set of content, typically 1-5% of the total volume, already flagged by machine learning. This enables them to focus on more valuable activities and still achieve comprehensive moderation coverage at a fraction of their existing cost. To set up human workforces and perform human review tasks, you can use Amazon Augmented AI, which is already integrated with Rekognition.

You can enhance the accuracy of the moderation deep learning model with the Custom Moderation feature. With Custom Moderation, you train a custom moderation adapter by uploading your images and annotating these images. The trained adapter can then be provided to the DetectModerationLabels operation to to enhance its performance on your images. See Enhancing accuracy with Custom Moderation for more information.

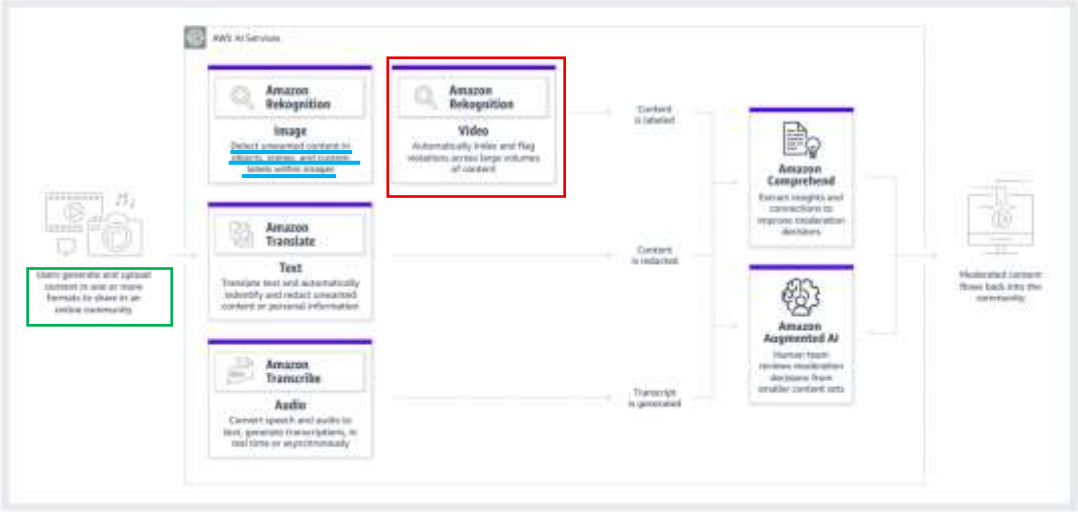
The following diagram shows shows the order for calling operations, depending on your goals for using the image or video components of Content Moderation:



Source: https://docs.aws.amazon.com/pdfs/rekognition/latest/dg/rekognition-dg.pdf#what-is

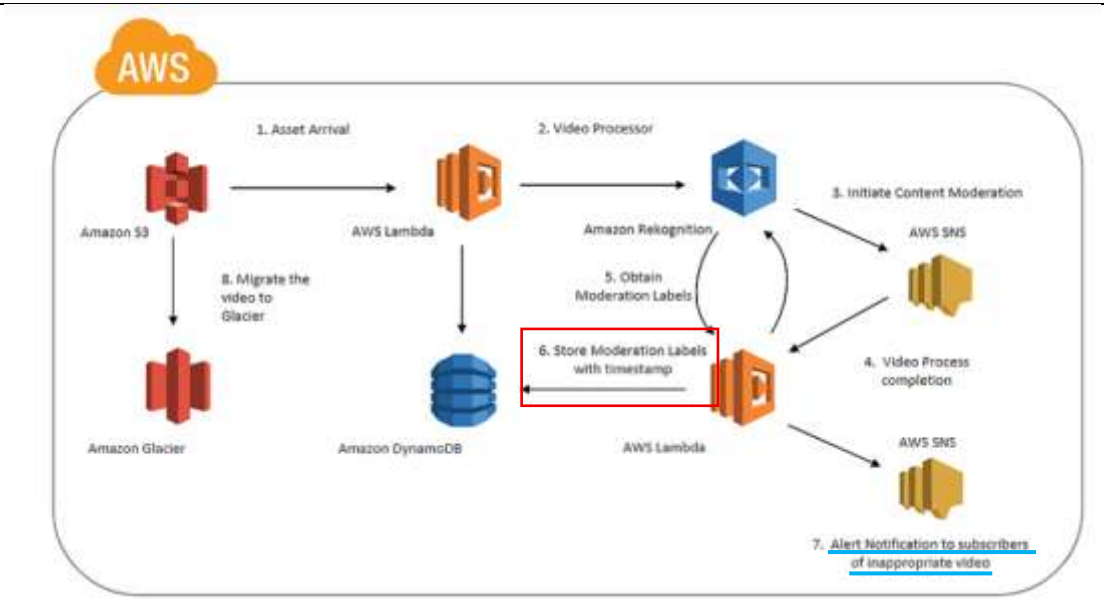
How it works

AWS fully managed AI services can work together to provide streamlined content moderation and provide total coverage of all content types. Increase moderation accuracy with additional contextual insights and bring human moderators in the loop to verify decisions when necessary and to improve the ML model over time.



Source: https://d1.awsstatic.com/psc-digital/2022/gc-400/ml-leading-use-case-ebook/204863_AWS_ML_Content_Moderation_eBook_Final.pdf

In this blog post, I show you how to build a serverless architecture to evaluate video content for enhanced compliance and moderation using Amazon Rekognition Video. Customers (especially media and entertainment companies) face the challenge of categorizing videos accordingly for building age-appropriate audience content or identifying inappropriate content in videos. Amazon Rekognition Video is a deep-learning-powered video analysis service that follows the path of objects, detects activities, and recognizes objects, celebrities, and content types. Amazon Rekognition Video can detect explicit and questionable content, so that you can filter videos based on your applications and your compliance requirements.

	 Processing starts immediately after media has been uploaded to our asset ingest Amazon S3 bucket. Amazon S3 event notifications triggers a video processor Lambda function that initiates the StartContentModeration API action on the video file. The completion status of media processing by Amazon Rekognition Video is obtained through Amazon SNS, which in turn triggers a content alert Lambda function. This Lambda function obtains the moderation labels by using the GetContentModeration API action, parsing the labels, and alerting through an SNS topic. Teams subscribed to the SNS topic will be immediately alerted with the name of the file and the count of moderation labels present. Complete moderation label information along with a timestamp is stored in the database. The video file in the S3 bucket is sent for archiving to Amazon Glacier, which provides low cost archive storage, and this completes our processing pipeline. Source: https://aws.amazon.com/blogs/machine-learning/build-an-automatic-alert-system-to-easily-moderate-content-at-scale-with-amazon-rekognition-video/
selecting for assessment the piece of digital content which caused the alert to be generated, wherein the piece of digital content is selectively modified;	*The method (e.g., Amazon Rekognition) further comprises selecting for assessment (e.g., moderating content) the piece of digital content which caused the alert to be generated (e.g., improper equipment, explicit and questionable content, etc.), wherein the piece of digital content is selectively modified.*

Automate moderation —

Enable human reviewers to follow up on smaller subsets of content, and protect them from harmful content exposure by automatically flagging up to 95% of unsafe content. Integrate human review without building new tools and infrastructure with Amazon Augmented AI (Amazon A2I).

Source: https://aws.amazon.com/rekognition/content-moderation/

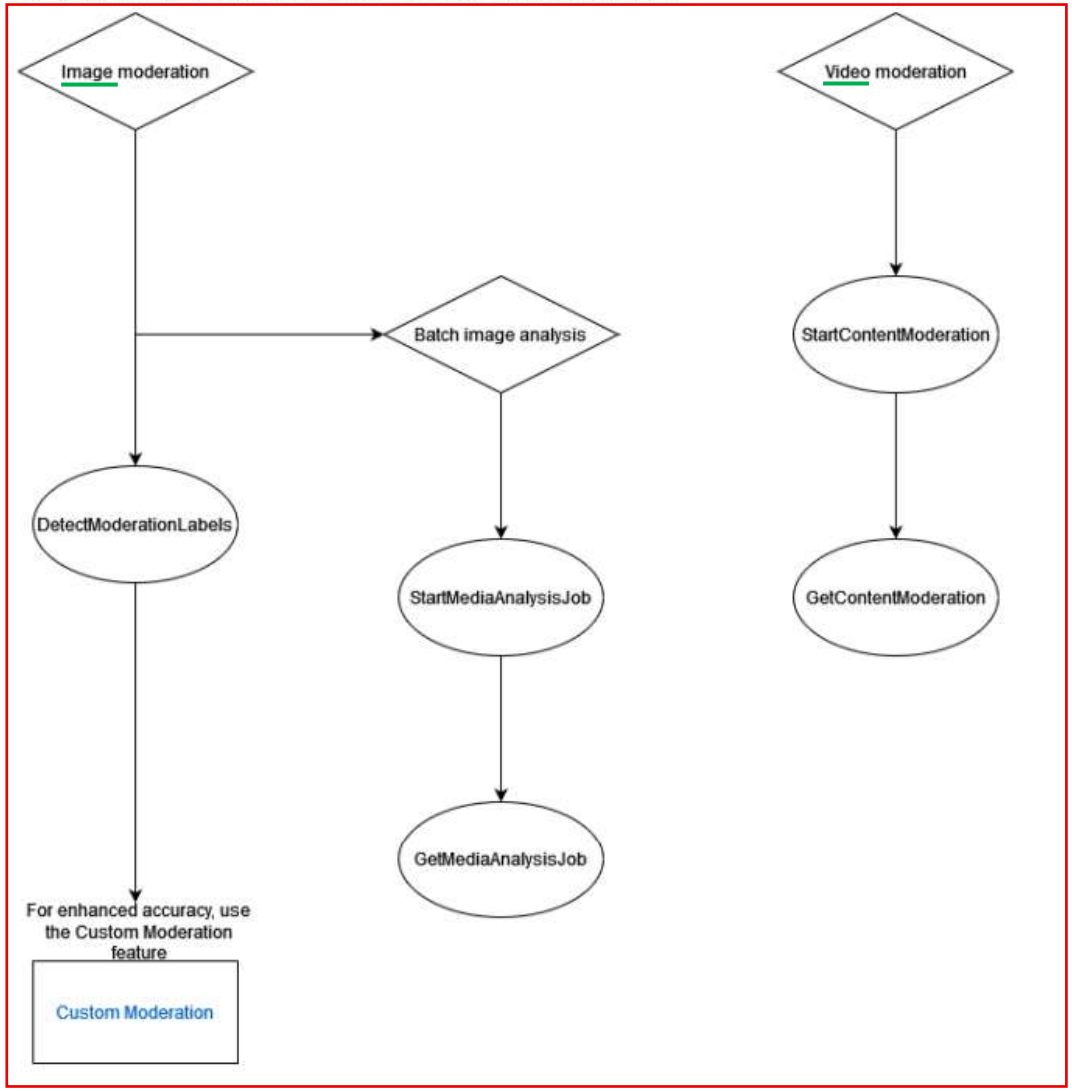
Moderating content

You can use Amazon Rekognition to detect content that is inappropriate, unwanted, or offensive. You can use Rekognition moderation APIs in social media, broadcast media, advertising, and e-commerce situations to create a safer user experience, provide brand safety assurances to advertisers, and comply with local and global regulations.

Today, many companies rely entirely on human moderators to review third-party or user-generated content, while others simply react to user complaints to take down offensive or inappropriate images, ads, or videos. However, human moderators alone cannot scale to meet these needs at sufficient quality or speed, which leads to a poor user experience, high costs to achieve scale, or even a loss of brand reputation. By using Rekognition for image and video moderation, human moderators can review a much smaller set of content, typically 1-5% of the total volume, already flagged by machine learning. This enables them to focus on more valuable activities and still achieve comprehensive moderation coverage at a fraction of their existing cost. To set up human workforces and perform human review tasks, you can use Amazon Augmented AI, which is already integrated with Rekognition.

You can enhance the accuracy of the moderation deep learning model with the Custom Moderation feature. With Custom Moderation, you train a custom moderation adapter by uploading your images and annotating these images. The trained adapter can then be provided to the DetectModerationLabels operation to to enhance its performance on your images. See Enhancing accuracy with Custom Moderation for more information.

The following diagram shows shows the order for calling operations, depending on your goals for using the image or video components of Content Moderation:



Using the image and video moderation APIs

In the Amazon Rekognition Image API, you can detect inappropriate, unwanted, or offensive content synchronously using DetectModerationLabels and asynchronously using StartMediaAnalysisJob and GetMediaAnalysisJob operations. You can use the Amazon Rekognition Video API to detect such content asynchronously by using the StartContentModeration and GetContentModeration operations.

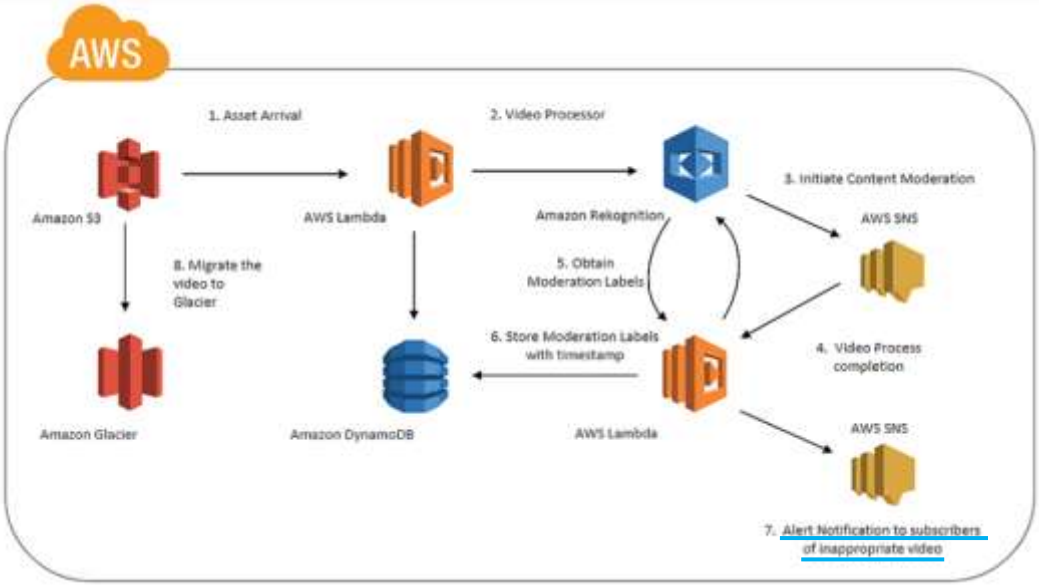
Label Categories

Amazon Rekognition uses a three-level hierarchical taxonomy to label categories of inappropriate, unwanted, or offensive content. Each label with Taxonomy Level 1 (L1) has a number of Taxonomy Level 2 labels (L2), and some Taxonomy Level 2 labels may have Taxonomy Level 3 labels (L3). This allows a hierarchical classification of the content.

For each detected moderation label, the API also returns the `TaxonomyLevel`, which contains the level (1, 2, or 3) that the label belongs to. For example, an image may be labeled in accordance with the following categorization:

L1: Non-Explicit Nudity of Intimate parts and Kissing, L2: Non-Explicit Nudity, L3: Implied Nudity.

Source: https://docs.aws.amazon.com/pdfs/rekognition/latest/dg/rekognition-dg.pdf#what-is



Processing starts immediately after media has been uploaded to our asset ingest Amazon S3 bucket. Amazon S3 event notifications triggers a video processor Lambda function that initiates the StartContentModeration API action on the video file. The completion status of media processing by Amazon Rekognition Video is obtained through Amazon SNS, which in turn triggers a content alert Lambda function. This Lambda function obtains the moderation labels by using the GetContentModeration API action, parsing the labels, and alerting through an SNS topic. Teams subscribed to the SNS topic will be immediately alerted with the name of the file and the count of moderation labels present. Complete moderation label information along with a timestamp is stored in the database. The video file in the S3 bucket is sent for archiving to Amazon Glacier, which provides low cost archive storage, and this completes our processing pipeline.

Example

For the purposes of the exercise, we are using a video from a swimming scene. When the video is processed the solution provides the following alert notification:

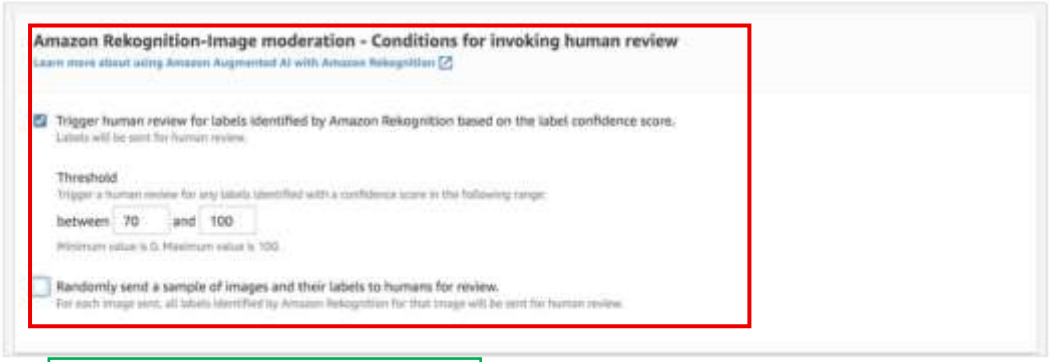


	Solution details
	
	Our Video Processor Lambda function is triggered by the asset's arrival in Amazon S3, and this triggers the Amazon Rekognition Video API to start content moderation. The JobId obtained for the analysis to be performed by Amazon Rekognition is stored in an Amazon DynamoDB table. The following code snippet from our Lambda function illustrates how we trigger Amazon Rekognition to initiate video content moderation and place the JobId information in a DynamoDB table:
	
	`response = rekognition.start_content_moderation(Video={"S3Object": {"Bucket": bucket,"Name": key}}`
	`NotificationChannel={"SNSTopicArn": snstopic,"RoleArn": rolearn},JobTag=jobtag)`
	
	`jobId=response["JobId"]`
	
	`table.put_item(Item={'jobid': jobId, 'videoname': key, 'Token':token, 'ContentModeration': 1})`
	
	Source: https://aws.amazon.com/blogs/machine-learning/build-an-automatic-alert-system-to-easily-moderate-content-at-scale-with-amazon-rekognition-video/

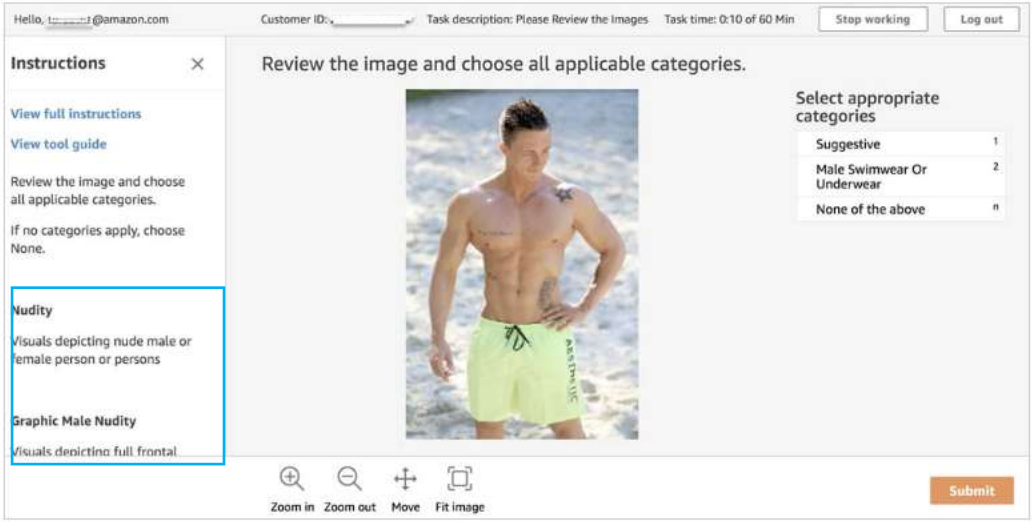
modifying metadata associated with the piece of digital content whether or not the piece of digital content is modified itself, based on one or both of the comparison which caused the alert and the modified piece of digital content; and	*The method (e.g., Amazon Rekognition) further comprises modifying metadata (e.g., "moderationLabelDetectionResult.getVideoMetadata();" etc.) associated with the piece of digital content (e.g., image or videos etc.) whether or not the piece of digital content is modified itself, based on one or both of the comparison which caused the alert (e.g., improper equipment, explicit and questionable content etc.) and the modified piece of digital content (e.g., moderated content).* Content moderation still requires human reviewers to audit results and judge nuanced situations where AI may not be certain in its prediction. Combining machine predictions with human judgment and managing the infrastructure needed to set up such workflows is hard, expensive, and time-consuming to do at scale. This is why we built Amazon Augmented AI (Amazon A2I), which lets you implement a human review of ML predictions and is directly integrated with Amazon Rekognition. Amazon A2I allows you to use in-house, private, or third-party vendor workforces with a web interface that has instructions and tools they need to complete their review tasks. You can easily set up the criteria that triggers a human review of a machine prediction; for example, you can send an image for further human review if Amazon Rekognition's confidence score is between 50–90. Amazon Rekognition handles the bulk of the work and makes sure that every image gets scanned, and Amazon A2I helps send the remaining content for further review to best utilize human judgment. Together, this helps ensure that you get full moderation coverage while maintaining very high accuracy, at a fraction of the cost to review each image manually.

7. In the **Amazon Rekognition-Image Moderation – Conditions for invoking human review** section, you can specify conditions that trigger a human review.

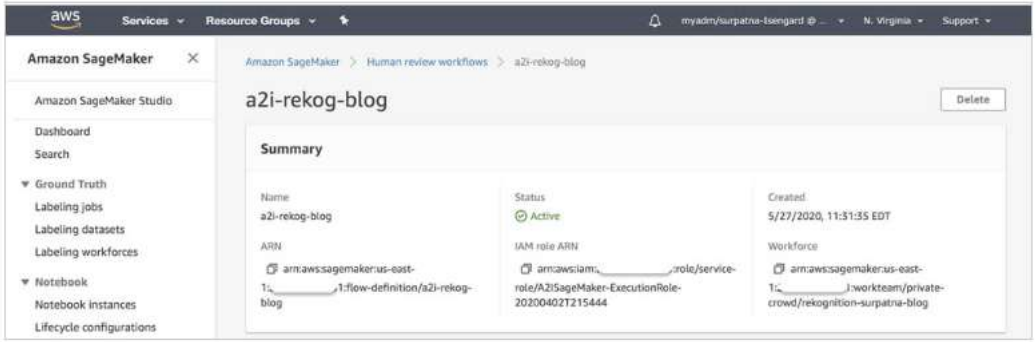
For example, if the confidence of the output label produced by Amazon Rekognition is between the range provided (70–100, for this use case), the document is sent to the portal for human review. You can also select different confidence thresholds for each image moderation output label through Amazon A2I APIs.



4. Choose an appropriate category for the image.



Before choosing **Submit**, if you go to the **Human review workflow** page on the Amazon SageMaker console and choose the human review workflow you created, you can see a **Human loops** summary section for that workflow.

	 5. In your worker portal, when you're done working, choose **Submit**. After you complete your job, the status of the human loop workflow is updated. If you navigate back to the **Human review workflow** page, you can see the human loop you just completed has the status `Completed`. Source: https://aws.amazon.com/blogs/machine-learning/build-alerting-and-human-review-for-images-using-amazon-rekognition-and-amazon-a2i/
compiling a digital data stream comprising: **any piece of digital content for which no alert was generated and its respective unmodified asset metadata describing that piece of unmodified content;** **any piece of digital content for which an alert was generated and which was not modified and its respective modified metadata**	*The method (e.g., Amazon Rekognition) further comprises compiling (aggregating) any piece of digital content for which no alert was generated and its respective unmodified asset metadata describing that piece of unmodified content (e.g., no moderation done on the segment), any piece of digital content for which an alert was generated and which was not modified (e.g., alert generated but no moderation done on the segment) and its respective modified metadata describing that piece of unmodified content and any piece of digital content for which an alert was generated and was modified (e.g., alert generated and moderation is done on the required segment etc.) and its respective modified metadata describing that piece of modified content.*

describing that piece of unmodified content; any piece of digital content for which an alert was generated and was modified and its respective modified metadata describing that piece of modified content; and	**Sorting and Aggregating** When retrieving results with GetContentModeration, you can sort and aggregate your results. **Sort order** — The array of labels returned is sorted by time. To sort by label, specify NAME in the SortByinput parameter for GetContentModeration. If the label appears multiple times in the video, there will be multiples instances of the ModerationLabel element. **Label information** — The ModerationLabels array element contains a ModerationLabel object, which in turn contains the label name and the confidence Amazon Rekognition has in the accuracy of the detected label. Timestamp is the time the ModerationLabel was detected, defined as the number of milliseconds elapsed since the start of the video. For results aggregated by video SEGMENTS, the StartTimestampMillis, EndTimestampMillis, and DurationMillis structures are returned, which define the start time, end time, and duration of a segment respectively. **Aggregation** — Specifies how results are aggregated when returned. The default is to aggregate by TIMESTAMPS. You can also choose to aggregate by SEGMENTS, which aggregates results over a time window. Only labels detected during the segments are returned. The evaluation manifest file contains: • Aggregated results as defined by F1Score • Details for the evaluation job including the ProjectVersionArn, number of training images, number of testing images, and the labels the adapter was trained on. • Aggregated TruePositive, TrueNegative, FalsePositive, and FalseNegative results for both base model and adapter performance. • Per label TruePositive, TrueNegative, FalsePositive, and FalseNegative results for both base model and adapter performance, calculated at the input confidence threshold. • Aggregated and per label TruePositive, TrueNegative, FalsePositive, and FalseNegative results for both base model and adapter performance at different confidence thresholds. The confidence threshold ranges from 5 to 100 in steps of 5.

	Video analysis With Amazon Rekognition Video, your applications can: • Track people and objects across video frames • Recognize objects • Recognize celebrities • Search stored and streaming video for persons of interest • Analyze faces for attributes like age and emotions • Detect explicit, inappropriate, or violent content or images • Aggregate and sort analysis results by timestamps and segments • Detect people, pets, and packages in streaming video Each video analysis operation returns metadata about the video being analyzed, as well as a job ID and a job tag. Operations like Label Detection and Content Moderation for video allow sorting by timestamp or label name, and aggregating results by timestamp or by segment.

Exercise 4: See aggregated metrics (console)

The Amazon Rekognition metrics pane shows activity graphs for an aggregate of individual Rekognition metrics over a specified period of time. For example, the `SuccessfulRequestCount` aggregated metric shows the total number of successful requests to all Rekognition API operations over the last seven days.

The following table lists the graphs displayed in the Rekognition metrics pane and the corresponding Rekognition metric. For more information, see CloudWatch metrics for Rekognition.

Graph	Aggregated Metric
Successful calls	SuccessfulRequestCount
Client errors	UserErrorCount

Graph	Aggregated Metric
Server errors	ServerErrorCount
Throttled	ThrottledCount
Detected labels	DetectedLabelCount
Detected faces	DetectedFaceCount

Each graph shows aggregated metric data collected over a specified period of time. A total count of aggregated metric data for the time period is also displayed. To see metrics for individual API calls, choose the link beneath each graph.

To allow users access to the Rekognition metrics pane, ensure that the user has appropriate CloudWatch and Rekognition permissions. For example, a user with `AmazonRekognitionReadOnlyAccess` and `CloudWatchReadOnlyAccess` managed policy permissions can see the metrics pane. If a user does not have the required permissions, when the user opens the metrics pane, no graphs appear. For more information, see Identity and access management for Amazon Rekognition.

For more information about monitoring Rekognition with CloudWatch see Monitoring Rekognition with Amazon CloudWatch.

To see aggregated metrics (console)

1. Open the Amazon Rekognition console at https://console.aws.amazon.com/rekognition/.
2. In the navigation pane, choose **Metrics**.
3. In the dropdown, select the period of time you want metrics for.
4. To update the graphs, choose the **Refresh** button.
5. To see detailed CloudWatch metrics for a specific aggregated metric, choose **See details on CloudWatch** beneath the metric graph.

The `GetLabelDetection` and `GetContentModeration` operations allow you to sort the analysis results by timestamp or by label name. You can also aggregate results by video segment or by timestamp.

To aggregate results by timestamp, set the value of the `AggregateBy` parameter to `TIMESTAMPS`. To aggregate by video segment, set the value of `AggregateBy` to `SEGMENTS`. `SEGMENTS` aggregation mode will aggregate the labels over time, while `TIMESTAMPS` gives the timestamp a label was detected at, using 2 FPS sampling and per frame output (Note: This current sampling rate is subject to change, assumptions shouldn't be made about the current sampling rate). If no value is specified, the default aggregation method is `TIMESTAMPS`.

```
#Copyright 2018 Amazon.com, Inc. or its affiliates. All Rights Reserved.
#PDX-License-Identifier: MIT-0 (For details, see https://github.com/awsdocs/
amazon-rekognition-developer-guide/blob/master/LICENSE-SAMPLECODE.)

    # ============== Unsafe content ===============
    def StartUnsafeContent(self):
        response=self.rek.start_content_moderation(Video={'S3Object': {'Bucket':
 self.bucket, 'Name': self.video}},
            NotificationChannel={'RoleArn': self.roleArn, 'SNSTopicArn':
 self.snsTopicArn})

        self.startJobId=response['JobId']
        print('Start Job Id: ' + self.startJobId)

    def GetUnsafeContentResults(self):
        maxResults = 10
        paginationToken = ''
        finished = False

        while finished == False:
            response = self.rek.get_content_moderation(JobId=self.startJobId,
                                                       MaxResults=maxResults,
                                                       NextToken=paginationToken,
                                                       SortBy="NAME",
                                                       AggregateBy="TIMESTAMPS")

            print('Codec: ' + response['VideoMetadata']['Codec'])
            print('Duration: ' + str(response['VideoMetadata']
['DurationMillis']))
            print('Format: ' + response['VideoMetadata']['Format'])
```

```
            print('Frame rate: ' + str(response['VideoMetadata']['FrameRate']))
            print()

            for contentModerationDetection in response['ModerationLabels']:
                print('Label: ' +
                    str(contentModerationDetection['ModerationLabel']['Name']))
                print('Confidence: ' +
                    str(contentModerationDetection['ModerationLabel']
['Confidence']))
                print('Parent category: ' +
                    str(contentModerationDetection['ModerationLabel']
['ParentName']))
                print('Timestamp: ' +
 str(contentModerationDetection['Timestamp']))
                print()

            if 'NextToken' in response:
                paginationToken = response['NextToken']
            else:
                finished = True
```

Source: https://docs.aws.amazon.com/pdfs/rekognition/latest/dg/rekognition-dg.pdf#faces

outputting the compiled digital data stream to a plurality of end users.

The method (e.g., Amazon Rekognition) further comprises outputting the compiled digital data stream (Aggregated data) to a plurality of end users.

Sorting and Aggregating Results

Results obtained from certain Amazon Rekognition Video operations can be sorted and aggregated according to timestamps and video segments. When retrieving the results of a Label Detection or Content Moderation job, with `GetLabelDetection` or `GetContentModeration` respectively, you can use the `SortBy` and `AggregateBy` arguments to specify how you want your results returned. You can use `SortBy` with `TIMESTAMP` or `NAME` (Label names), and use `TIMESTAMPS` or `SEGMENTS` with the `AggregateBy` argument.

Source: https://docs.aws.amazon.com/pdfs/rekognition/latest/dg/rekognition-dg.pdf#faces

Disclaimer

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